

Week 4 — `week04_inference.ows`

Companion notes for the Orange workflow. Book: chapter 4. Notebook twin: `notebooks/week04_inference.ipynb`.

1. What this workflow is

Five visual widgets that show **shape** and **association**, one Select Rows branch that isolates a single segment, and one Python Script that performs every **test** Orange has no widget for.

File: <code>ab_test.tab</code>	→ Feature Statistics	§ 2
	→ Distributions	§ 2, § 0
	→ Box Plot (prints a t-test)	§ 5
	→ Sieve Diagram	§ 1, § 5
	→ Mosaic Display	§ 7
	→ Select Rows → Sieve (one segment)	§ 6
	→ Python Script → Data Table	§ 6, § 7

The § references point at the matching section of the notebook, which follows chapter 4's own nine-section order.

Dataset — `data/ab_test.tab`, 70,000 sessions from an A/B test. `variant` (A/B), `device` (mobile/desktop), `day` (1–14), `session_seconds`, `pages_viewed`, and the target `converted` (0/1).

The idea being taught: know the limits of your tool. Orange is a visual exploration environment, not a statistics package. It can show you that two bars differ; it will not hand you a z-statistic, a p-value or a confidence interval for a two-proportion test. So the workflow does what a real analyst does — it uses the visual widgets to *find* the effect and drops into Python to *quantify* it.

Pre-configured: the Python Script widget already contains the analysis, so the workflow runs as shipped. The same script lives at `orange/scripts/ab_test.py` and runs standalone with `python orange/scripts/ab_test.py`, which is how it is tested.

2. What to do

Shape — what kind of variable is each column?

1. **Feature Statistics** — read the thumbnail histogram beside each column. Which are symmetric and which have a long right tail?
2. **Distributions** — `pages_viewed`. A count of events in one visit: that is a Poisson shape. Then `session_seconds`: the long right tail is log-normal.
3. **Distributions** — `converted`, split by `variant`. Eyeball the difference. Is it convincing?

Association — do two variables move together?

1. **Box Plot** — `session_seconds` by `variant`. Orange prints a t-test at the bottom of this widget; it is the only test it performs for you.
2. **Sieve Diagram** — `variant` × `converted`. Colour shows the deviation from independence; grid density shows the cell frequency.
3. **Mosaic Display** — `variant`, `converted` and `device`. Two questions: does the lift hold in both devices, and is the device *mix* the same in both arms? The second decides whether `device` could confound anything at all.

One segment at a time

1. **Select Rows** — set the condition `device = mobile` and read the Sieve downstream. Then switch it to `desktop` and read it again. That is two hypothesis tests, chosen after seeing the data.

The numbers Orange will not give you

1. **Python Script** — run it. It prints, in the order a real analysis runs: the sample-ratio check, the lift with a 95% interval, the sample size that was actually required, and the per-device segments with a Holm correction.

Versi mahasiswa. Pembahasan, hal yang perlu diperhatikan, dan jawaban setiap langkah disampaikan di kelas setelah praktikum.